John Zhang

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TECHNICAL SKILLS

Languages: Python, C++, C, Rust, TypeScript, JavaScript, Java
ML/Infra: PyTorch, TensorFlow, JAX, CUDA, TensorRT, NumPy, Pandas, Keras, scikit-learn, matplotlib, wandb, Docker
Robotics/CV: ROS2, Jetson Orin Nano, VLMs, OpenCV, YOLO, MediaPipe, SLAM, Isaac Sim, MuJoCo, Linux,

EXPERIENCE

Machine Learning Engineer May 2026 – NOW
Mecka AI *Toronto, ON*

- Led v2 development of internal **PII-Blur model and pipeline** that all company deliveries run on, improving **recall 53.9%** ⇒ **85.7%** and **precision 69.7%** ⇒ **87.4%**, reducing false positives by 25.4%.
- For EgoExo project, integrated internal gapfiller model, VGGT, calibration pipeline v2, multi-person selection, cpu stage optimizations **(1.22x)** and **improved CV loss by 40ish%.**
- Researched and experimented with VLM **test-time RL** methods using GRPO for Mecka's SOTA video labeling system .
- Fine-tuned labeler using DPO with preference data augmentation and hyperparameter tuning, improving from **72.4%** to **79.2%** accuracy scored by internal Qwen verifier and human audit.
- Trained impossible hand pose **classification model** for Mecka's CV hands pipeline, MLP on 10 hours of **Manus sensor gloves** data, achieving **89.4%** accuracy scored against pelvis and detected hands divergence.
- Developed Mecka's MoCap sync pipeline from scratch, between **OptiTrack** data, ego video with DS calibration, and **Manus 25-DOF** sensor gloves. Used for hands and UMI gripper ground truth for CV evaluations.

Machine Learning Engineer May 2025 – Dec 2025
WAT.ai *Waterloo, ON*

- Built an emotion analysis system on the **BoLD** body-language dataset, preprocessing video clips into per-frame pose features.
- Fine-tuned **YOLOv8** on **COCO** joint-keypoint data for expression and pose detection, achieving **82.2%** cross-domain accuracy on BoLD.
- Researched and benchmarked **TransVOD-DETR** for video emotion recognition on BoLD, improving **multi-frame F1 by 12%** over the single-frame baseline.

Research Assistant Jan 2025 – May 2025
Alberta Machine Intelligence Institute (Amii), supervised by Prof. Grzegorz Kondrak & Ning Shi (Ph.D.) *Remote*

- Coauthor, Paper **accepted at EACL 2026**; contributed to inference logic script, case studies, and related work analysis on contradiction detection with word surprisal.
- Coauthor, Paper **accepted at SemEval 2025 EA-MT**; contributed to paper writing, methods design, experiments evaluation for Entity-Aware Machine Translation task.
- Achieved **1st / 27 research teams** in **SemEval 2025 EA-MT** for Arabic translation and 4th overall.
- Reviewed papers as *secondary reviewer for **EMNLP 2025** and **COLM 2025** submissions on VLMs and semantic change detection*
- Integrated transformers, named-entity recognition and RAG modules for +**10% in benchmark scores** on validation sets.
- Developed PyTorch inference pipeline to extract next-word probabilities from token-level logits in byte-level LLMs.

PROJECTS

Simulated Autonomous Navigation System | *C++, ROS2, Foxglove, Gazebo, Docker* May 2025

- Implemented full ROS 2 navigation stack in Gazebo, including costmap generation, global map fusion, and **A* path planning**.
- Wrote Costmap, Map Memory, Planner, and Control nodes for real-time map fusion, A* planning, and Pure Pursuit control.
- Validated the stack in **Foxglove** to ensure stable point-to-point navigation and collision-free behavior in simulation.

Gesture OS | *Python, TensorFlow, OpenCV, NumPy, Keras, MediaPipe* *April 2025*

- Trained two small feedforward MLPs in TensorFlow using Mediapipe gesture dataset to classify spatial gestures and temporal tracking.
- Used the classifier models to develop a real-time computer vision system enabling gesture-based device control using OpenCV and MediaPipe, getting **100% accuracy** in recorded hand landmark detection and gesture tracking.

EDUCATION

University of Waterloo Waterloo, ON
B.A.Sc. in Computer Engineering | CGPA: 3.8/4.0 | President's Scholarship of Distinction ($2000) *Expected Apr 2028*